AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX EXAMINATION

MAY 2013

Physics Paper II

Time: 2 hours 25 minutes Marks: 40

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are NINE questions. Answer ALL questions. Questions 7, 8 & 9 each offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

   Read each question carefully.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a simple calculator if you wish.

Q.1. (Total 3 Marks)

"Physics helps in attaining sustainable development of our society." Discuss the given statement by mentioning THREE important roles of physics.

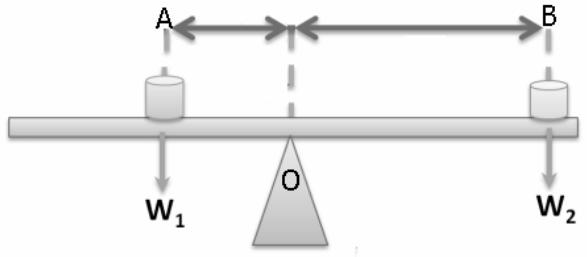
Q.2. (Total 6 Marks)

a. Explain that motion and rest are relative terms. (3 Marks)

b. Give any THREE examples to explain how a body can be at motion and in rest simultaneously. (3 Marks)

Q.3. (Total 3 Marks)

A metre rod is supported at its middle point O as shown in the given figure.



A block of weight 5 N is suspended at point B, 20 cm from O. Find the weight of the block that balances it at point A, 10 cm from O.

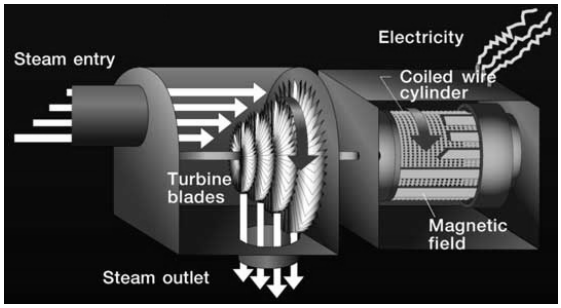
Q.4. (Total 3 Marks)

Two marble balls of 36 kg and 20 kg are placed in such a way that the distance between their centres is 60 cm. Find the force of attraction between them.

(Take $G = 6.67 \times 10^{-11} Nm^2/kg^2$).

Q.5. (Total 6 Marks)

a. The given diagram represents a steam-powered generator. Describe how energy is transformed from one form to another. (4 Marks)



b. Write TWO drawbacks of this process of power generation associated with the environment.

(2 Marks)

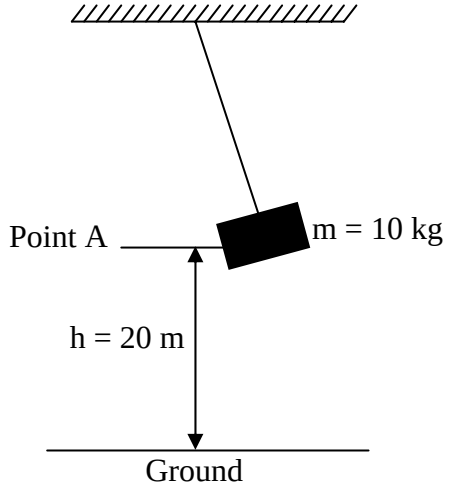
Q.6. (Total 4 Marks)

A person is sitting in front of a room heater. Explain the possible ways through which heat is transferred to that person.

Q.7. (Total 5 Marks)

EITHER

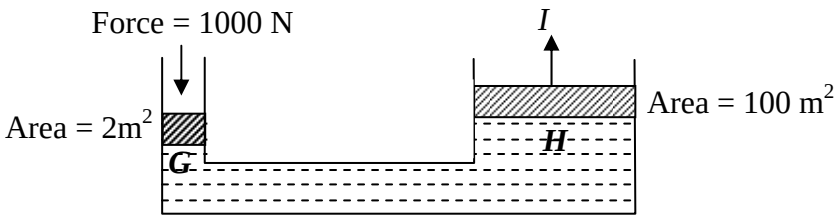
a. The diagram shows a block hanging by means of a fixed support.



Calculate the potential energy (P.E) of the block at point A with respect to ground. If the block is released, calculate the energy with which it strikes the ground.

OR

b. The given figure shows the brake system of a vehicle.



Name the law which is used in the above system. Calculate the pressure exerted by the liquid at *G* and *H* and the force which is applied at *I*.

Q.8. (Total 5 Marks)

EITHER

a. A boy walks from 750 m North and then 500 m South-East at 40˚ to reach home.
Draw a vector diagram to determine the resultant displacement of the boy by using a suitable scale.

Calculate the velocity of the boy if he takes 5 minutes to reach home.

OR

b. A force of 40 N is applied to a carton in rightward direction on a frictionless horizontal surface to produce an acceleration of 3 m/s^2.

Show the direction of force and acceleration in the given diagram and calculate the mass of the block.



Q.9. (Total 5 Marks)

EITHER

a. Describe any FIVE factors which affect surface evaporation.

OR

b. Describe any FIVE uses of good and bad conductors.

END OF PAPER

Please use this page for rough work